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ORIGINAL PAPER

Robust estimation of Markov-switching GARCH models

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Abstract
The study of volatility is important in several areas of finance, and GARCH models have been widely used in the literature due to their ability to capture key stylised facts of financial time series. However, in some cases, financial time series exhibit structural changes in volatility dynamics, for which standard GARCH models may be inadequate; in such situations, Markov-switching GARCH models provide a more suitable framework. On the other hand, outliers are often present in empirical data. This paper shows that conventional estimators can be strongly affected by outliers and proposes an estimator that is more robust to their presence. The simulation results and empirical applications indicate that the proposed robust estimator is competitive and useful under contamination, but not uniformly superior to the QML-t estimator.

Keywords Robust statistical methods · Volatility models · Hidden Markov models · Outliers · Value at risk

1 Introduction

Volatility has been one of the most intensively studied aspects of financial econometrics in recent decades. From an empirical perspective, several authors have documented that financial time series exhibit a number of stylised facts (Francq and Zakoian 2019, Section 1.3, p. 7). In this context, Engle (1982) proposed the autoregressive conditional heteroskedasticity (ARCH) model. However, in many

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applications, the estimated ARCH models require high orders. To address this issue, Bollerslev (1986) introduced an extension of the ARCH model, namely the Generalised ARCH (GARCH) model.

The most popular GARCH estimator is the quasi-maximum likelihood (QML) estimator. Francq and Zakoian (2019, Section 7, p. 175) showed that, under certain conditions, this estimator has excellent asymptotic properties, such as consistency. However, this estimator is not robust to outliers.

Two approaches are frequently used to deal with outliers: the first is to detect outliers and treat them in some appropriate way; and the second is to use estimators that are robust to outliers. Hotta and Tsay (2012), based on ARCH(p) and GARCH(1,1) models, studied the impact of outliers on maximum likelihood estimation as well as proposing a method for detecting outliers. Several authors have presented robust methods to estimate GARCH models. For example, Sakata and White (1998) used S-estimators; Muler and Yohai (2008) presented an estimator based on M-estimators combined with a robust filter mechanism to reduce the effect of outliers on conditional variances, and Boudt et al. (2013) proposed a modification of the objective function of the M-Estimator and the mechanism used by Muler and Yohai (2008). In the presence of outliers these estimators have a significant gain over the non-robust estimators, as shown by Carnero et al. (2012) for the estimator of Muler and Yohai (2008). Additionally Trucíos et al. (2017) presented a modification for robust prediction of volatility in GARCH models.

On the other hand, financial time series often exhibit structural breaks over time, usually associated with political events and/or financial crises, such as the subprime crisis of 2008–2009. Consequently, especially in long time series, it is inappropriate to assume that a single set of GARCH parameters governs the entire sample period. In the presence of structural breaks, such models may become inadequate. Hillebrand (2005) pointed out that this adverse effect is independent of the estimation method. For instance, Hwang and Valls Pereira (2008) showed that the estimated persistence of conditional volatility in large samples can be overstated, particularly when structural breaks affect the ARCH and GARCH parameters.

Hamilton (1989) and Cai (1994) proposed the conditional heteroscedastic autoregressive model with Markov-switching (MS-ARCH) regime. By allowing switching between different regimes, this model is able to capture more complex dynamic patterns. Because the regimes are not observable, in the maximum likelihood estimation of the model, one needs to infer the probability of being in a given regime in time. For this, the mechanism used is the Hamilton (1989) filter.

When extending this method to GARCH Markov-switching (MS-GARCH) models, it was observed that the maximum-likelihood estimation was unfeasible because it depended on all possible path combinations. This problem can be circumvented via Bayesian estimation, as shown by Das et al. (2004) and Bauwens et al. (2010). In the traditionl approach, Gray (1996) was the first author who dealt with the path dependency problem making maximum likelihood estimation possible. Later, Haas et al. (2004) presented a new approach to MS-GARCH models, with a new interpretation of the model presented by Gray (1996).

MS-GARCH models have become popular in the literature because, in many cases, they outperform single-regime GARCH models. For instance, Ardia et al. (2018)

found that MS-GARCH models provide better predictions of VaR and expected shortfall compared to GARCH models with a single-regime in several settings. Furthermore, they observed that the improvements were more pronounced when the Markov switching mechanism was applied to simpler specifications, such as the GARCH-Normal model.

However, like other standard filters, Hamilton (1989)'s filter performs poorly in the presence of outliers as shown by Calvet et al. (2015). The objective of this paper is therefore to develop robust alternatives to maximum likelihood estimation for MS-GARCH models. To this end, we propose several modifications, including the use of robust filters as in Petrus (1999) and Calvet et al. (2015), guided by the principles of robustness outlined in Huber and Ronchetti (2009, Section 1.2, p. 5). In particular, an estimator should: (i) maintain reasonably good efficiency under the assumed model; (ii) be robust in the sense that small deviations from the model assumptions have only a limited impact on performance; and (iii) remain stable under moderate departures from the model, avoiding catastrophic failures. In this spirit, we introduce further modifications to the estimation procedures for MS-GARCH models.

As far as we know, there is no work in the literature dealing with the robust estimation MS-GARCH models. The goal of the paper is to show the effect of outliers on the estimation of MS-GARCH models and then to propose a robust alternative. In Sect. 2, the GARCH model and non-robust and robust estimation methods are presented, in particular the robust methods of Muler and Yohai (2008) and Boudt et al. (2013). In Sect. 3, MS-GARCH models are presented along with their properties, interpretation and advantages in relation to GARCH models in certain scenarios. Among the different specifications of MS-GARCH models, we present the model of Haas et al. (2004). In addition, we describe an alternative robust estimation that consists of adapting the techniques used by Boudt et al. (2013) to the case of regime-switching models. Section 4 presents a Monte Carlo simulation study to evaluate the performance of the estimators of the MS-GARCH models. We evaluate both the quasi-likelihood estimation through Hamilton (1989)'s filter and robust alternatives. In this section we focus on evaluating the accuracy of the parameter estimates and the ability of the filter to identify the regimes. We observe that, in general, the QML estimators using the Student-t (QML-t) and the robust Hamilton filter perform reasonably well in identifying regimes, unlike the QML using the normal distribution (QML-n). On the other hand, when estimating the conditional variance parameters, we note that the estimators tend to estimate some parameters well, but not others. Applications to four exchange rate series are presented in Sect. 5. Finally, in Sect. 6 the main conclusions found in the simulations and applications in relation to the considered estimators are discussed.

2 GARCH models

According to Bollerslev (1986), y_t follows a GARCH(p,q) model if:

$$
\begin{aligned}
y_t &= \sqrt{h_t} z_t, \\
h_t &= \alpha_0 + \sum_{i=1}^{p} \alpha_i y_{t-i}^2 + \sum_{j=1}^{q} \beta_j h_{t-j},
\end{aligned}
\tag{1}
$$

where $\{z_t\}$ is a sequence of independent and identically distributed (i.i.d.) random variables with mean 0 and variance 1; and $\alpha_0 > 0$, $\alpha_i \geq 0$, and $\beta_j \geq 0$ are the sufficient conditions for h_t to be positive. Stationary conditions can be found, for instance, in Francq and Zakoian (2019, Section 2.2.2) and in the Supplementary Materials.

2.1 Estimators

Various methods exist to estimate GARCH models. For instance, Baillie and Chung (2001) estimated the GARCH(1,1) model using a minimum distance estimator for the squared process autocorrelations, while Pérez-Cruz et al. (2003) used the support vector machine method. Aknouche and Guerbyenne (2006) developed three recursive online algorithms based on a two-stage least squares scheme, and Ardia (2008, Section 3) presented methods for Bayesian estimation. From a frequentist perspective, the most commonly used approach is QML estimation.

2.1.1 Quasi-maximum likelihood estimator

Denote $\boldsymbol{y}_T = \{y_1, y_2, \ldots, y_T\}$, $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \boldsymbol{\beta}')'$, with $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \ldots, \alpha_p)'$ and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_q)'$, so that it belongs to the parametric space of the form $\Theta \subset (0, +\infty) \times [0, \infty)^{p+q}$ - often the space in which the GARCH model is stationary. To obtain the likelihood of the GARCH(p,q) model defined in (1), it is necessary to specify the distribution of $\{z_t\}$. The name quasi-likelihood comes from the fact that one distribution is used, even when knowing that this distribution is not correct. For the normal case, given the initial values, for example, $\boldsymbol{\phi}_0 = \{y_0, \ldots, y_{1-p}, h_0, ..., h_{1-q}\}$, the quasi-likelihood can be expressed as:

$$
L\left(\boldsymbol{\theta} | \boldsymbol{y}_T, \boldsymbol{\phi}_0\right) = \prod_{t=1}^{T} f\left(y_t | \mathcal{I}_{t-1}, \boldsymbol{\theta}, \boldsymbol{\phi}_0\right) = \prod_{t=1}^{T} \frac{1}{\sqrt{2\pi h_t}} \exp\left\{-\frac{y_t^2}{2h_t}\right\}, \tag{2}
$$

where h_t can be obtained recursively through (1). Thus, the QML estimate $\hat{\boldsymbol{\theta}}$, assuming the standard normal distribution (QML-n), is given by:

$$
\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta} \in \Theta}{\arg\max}\, L\left(\boldsymbol{\theta} | \boldsymbol{y}_T, \boldsymbol{\phi}_0\right). \tag{3}
$$

Or by maximising the log-likelihood:

$$
\ell\left(\boldsymbol{\theta} | \boldsymbol{y}_{\boldsymbol{T}}, \boldsymbol{\phi}_0\right) = \log\left(L\left(\boldsymbol{\theta} | \boldsymbol{y}_T, \boldsymbol{\phi}_0\right)\right) \propto -\frac{1}{2} \sum_{t=1}^{T} \left[\ln(h_t) + \frac{y_t^2}{h_t}\right]. \tag{4}
$$

Under certain conditions, the QML estimator is consistent, and the distribution converges (Francq and Zakoian 2019, Sect. 7.1.1), i.e.,

$$\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) \xrightarrow{\mathrm{L}} N\left(0, (\kappa_z - 1)\, J^{-1}\right), \quad \text{where}$$

$$\kappa_z = E(z_t^4) \quad \text{and} \quad J := E\left(\frac{\partial^2 l(\boldsymbol{\theta}|\boldsymbol{y_t}, \phi_0)}{\partial \boldsymbol{\theta}\boldsymbol{\theta}'}\right).$$

The choice of initial values in finite samples can produce different values of the estimates, especially when the sample size is small. Pelagatti and Lisi (2009) showed the accuracy of the QML-n estimates for the GARCH(1,1) model under different initial value choices.

2.2 Robust estimation of GARCH models

There are numerous methods to robustify estimators of GARCH models, focusing mainly on additive outliers. We highlight Basu et al. (1998), Sakata and White (1998), Park (2002), Lee and Song (2009), Hotta and Trucíos (2018) and Crosato and Grossi (2019). However, Muler and Yohai (2008) pointed out that the robust estimators described above may still be sensitive to a fraction of outliers in the sample because they rely on the conditional variance, which is itself highly sensitive to outliers. To address this limitation, they proposed a modification of the M-estimators together with a modified specification of the GARCH conditional variance aimed at reducing the influence of outliers.

2.2.1 BIP-GARCH model

Muler and Yohai (2008) proposed a mechanism similar to robust filtering to estimate the conditional variance. This mechanism, called bounded innovation propagation (BIP), restricts the propagation of the outliers' effect present at time $t-1$ on the conditional variance of time t. For the GARCH(p,q) model, h_t is calculated as:

$$\begin{aligned} h_t^* &= \alpha_0 + \sum_{i=1}^{p} \alpha_i h_{t-i}^* r_k\left(\frac{y_{t-i}^2}{h_{t-i}^*}\right) + \sum_{i=1}^{q} \beta_j h_{t-i}^*, \\ r_k(u) &= \begin{cases} u, & \text{if } u \le k \\ k^*, & \text{if } u > k, \end{cases} \end{aligned} \tag{5}$$

with $k^* = k$. However, Boudt et al. (2013) observed in their simulations that, when using the BIP specification in (5), the parameter estimates of the GARCH model exhibit small biases. To mitigate this problem, they introduced a correction factor, c_{k_δ}, such that the conditional expectation of $c_{k_\delta} h_{t-1}^{(k)} r_{k_\delta}(y_{t-1}^2/h_{t-1}^{(k)})$, given $\mathcal{I}_{t-2}$, is equal to the conditional expectation of y_{t-1}^2 in the absence of outliers. Here, $\mathcal{I}_{t-2}$ denotes the information set available up to time $t-2$. Consequently, Eq. (5) becomes

$$
\begin{aligned}
h_t^* &= \alpha_0 + \sum_{i=1}^{p} \alpha_i h_{t-i}^* c_{k_\delta} r_{k_\delta}\left(\frac{y_{t-i}^2}{h_{t-i}^*}\right) + \sum_{i=1}^{q} \beta_j h_{t-i}^*, \\
r_{k_\delta}(u) &= \begin{cases} u, & \text{if } u \leq k_\delta \\ k_\delta^* & \text{if } u > k_\delta. \end{cases} \\
c_{k_\delta} &= \frac{E[W]}{E\left[r_{k_\delta}(W)W\right]} = \frac{1}{F_{\chi_3^2}(k_\delta) + (1-\delta)k_\delta},
\end{aligned}
\tag{6}
$$

where $k_\delta^* = k_\delta$, and W is a chi-square random variable with one degree of freedom, $F_{\chi_3^2}(.)$ is the cumulative distribution function of the chi-square distribution with three degrees of freedom and k_δ is the δ quantile of the chi-square distribution with one degree of freedom. The notation k_δ was introduced to clarify the meaning of the threshold, from which one suspects that the observed value is affected by an outlier. Table 1 of the article by Boudt et al. (2013) presents the values of c_{k_δ}.

Other authors have proposed different values of k^* and k_δ^* in (5) and (6), respectively. Carnero et al. (2012) proposed using $k^* = 1$. i.e., when there is an atypical observation, it is replaced by $E[y_{t-1}^2] = h_{t-1}$. For cases where another value for k_δ is used in (6), when $u > k_\delta$, say k_{δ^*}, we have that the value of c_{k_δ} is given by $c_{k_\delta} = [F_{\chi_3^2}(k_\delta) + k_{\delta^*}(1 - F_W(k_\delta))]^{-1}$. For the case of Carnero et al. (2012), where it is replaced by 1, we have that $\delta^* = 0.6827$.

The constant δ acts as a trade-off between robustness and efficiency. If the chosen value is too high h_t^*, tends to coincide with h_t, whereas for smaller values, h_t^* becomes more robust to outliers.

Boudt et al. (2013) proposed to use variance target, that is, α_0 is reparameterized as $\hat{h}(1 - \alpha_1 - \beta_1)$—with $\hat{h}$ being a consistent estimator of the unconditional variance h is estimated before the other parameters. In the absence of outliers, $\hat{h}$ can be obtained simply from the sample variance; however, this estimator is highly sensitive to the presence of outliers. A robust alternative for estimating the unconditional variance is the reweighted variance estimator proposed by Boudt et al. (2013). The reweighted variance estimator, when $z_t \sim N(0, 1)$, is given by:

$$
\begin{aligned}
\hat{h} &= 1.318 \frac{\sum_{t=1}^{T} y_t J_t}{\sum_{t=1}^{T} J_t} \\
J_t &= I\left[\frac{y_t^2}{\text{MAD}_t^2} \leq \chi_1^2(95\%)\right],
\end{aligned}
$$

such that $I(.)$ is the indicator function and the MAD of a sequence of observations $y_1, \ldots, y_T$ is defined as $1.486 \times \text{median}_i(|y_i - \text{median}_j(y_j)|)$, where 1.486 is a correction factor to ensure that the MAD is a consistent scale estimator of the normal distribution.

Finally, Boudt et al. (2013) estimated the parameters of the BIP-GARCH model using an estimator that minimises an objective function ρ evaluated for the logarithm of the squared standardised observations. This is,

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}\in\Theta} \frac{1}{T}\sum_{t=1}^{T} \rho\left(\ln \frac{y_t^2}{h_t}\right). \tag{7}$$

Based on the comparison of several ρ candidate functions, Boudt et al. (2013) recommended one associated with the Student-t density with four degrees of freedom. In the more general case of a Student-t density with with ν degrees of freedom, the corresponding ρ-function is given by

$$\rho(z) = -z + \sigma_\nu \rho_{t_\nu}(\exp(z)), \tag{8}$$

$$\rho_{t_\nu}(u) = (1+\nu)\log\left(1 + \frac{u}{\nu-2}\right), \tag{9}$$

$$\sigma_\nu = \frac{1}{E[\rho'_{t_\nu}(W)W]}. \tag{10}$$

2.3 Volatility prediction

For simplicity of exposition and notation, we consider the GARCH(1,1) model defined in (1) to illustrate the computation of the τ-step-ahead volatility forecast. Suppose the model has been estimated over the time period $t = 1, 2, \ldots, T$. The τ -steps-ahead volatility forecast at T, given the information available up to this time, denoted by $h_{T+\tau|T}$, and minimising the mean squared error, is given by

$$h_{T+\tau|T} = \alpha_0 + \alpha_1 y^2_{T+\tau-1|T} + \beta_1 h_{T+\tau-1|T}, \tag{11}$$

where, similarly, $y^2_{T+\tau|T}$ is the forecast of y_t^2 τ-steps ahead, conditional on $\mathcal{I}_T$; that is, $y^2_{T+\tau|T} \equiv E[y^2_{T+\tau}|\mathcal{I}_T]$. $\boldsymbol{\theta} = (\alpha_0, \alpha_1, \beta_1)'$ can be estimated by (3). As with parameter estimation, the presence of outliers affects the volatility forecast. To deal with this, an alternative to the forecast τ-steps-ahead of the conditional variance, as presented by Boudt et al. (2013), is given by:

$$h_{T+\tau|T} = \alpha_0 + \alpha_1 c_{k_\delta} h_{T+\tau-1|T} r_{k_\delta}\left(y^2_{T+\tau-1|T}/h_{T+\tau-1|T}\right) + \beta_1 h_{T+\tau-1|T}, \tag{12}$$

where $r_{k_\delta}(.)$ is the function defined in (6). As noted by Boudt et al. (2013), in the presence of outliers, y_t^2 is not observed and is replaced by h_t when using the r_{k_δ} mechanism. Thus, the effect of these outliers on future variations is unlimited in (11), but limited in (12). Additionally, Carnero et al. (2012) observed that, in the presence of outliers, non-robust estimators of the parameters of the GARCH model generally lead to biases in the estimation of volatilities.

3 Markov-switching GARCH models

Lamoureux and Lastrapes (1990), Horváth et al. (2006) and other authors have provided empirical evidence of structural changes of stock returns. In this scenario, as cited by Gray (1996), many popular models produce poor results when fitted to the data. Lamoureux and Lastrapes (1990), Hillebrand (2005), Hwang and Valls Pereira (2008) and other authors have shown that in the presence of structural breaks in the parameters of the ARCH and GARCH processes, when adjusting the GARCH model, the estimated persistence, even in large samples, may be excessive. Furthermore, Hillebrand (2005) stated that this negative effect is independent of the estimation method of the GARCH model. Additionally, some authors have observed this fact empirically, such as Mikosch and Stărică (2004), who estimated a GARCH model in a sample that exhibits structural changes in its conditional variance and obtained persistence practically equal to that of Billio et al. (2016).

To deal with this situation, Cai (1994) and Hamilton and Susmel (1994) proposed MS-ARCH models. The idea of the MS approach to modelling heteroscedasticity is that the distribution of the ARCH process depends on an unobserved state. The MS-ARCH model presented by Cai (1994), for example, is given by:

$$\begin{aligned} y_t &= \sqrt{h_t} z_t, \\ h_t &= \alpha_0^{(\Delta_t)} + \sum_{i=1}^{p} \alpha_i y_{t-i}^2, \end{aligned}$$

where Δ_t is a homogeneous ergodic Markov chain on a finite set $S = \{1, \ldots, K\}$.

However, extending this specification to the MS-GARCH framework is not feasible. Due to the path dependence of the model, the likelihood function for a sample size T requires integration over all K^T possible combinations up to the T-th observation, making direct likelihood evaluation computationally impracticable.

Gray (1996) was the first to address the path dependence problem in MS-GARCH models. He showed that the conditional density of y_t can be expressed as a mixture of normal distributions, with time-varying mixture weights $p_{t-1}(\Delta_t = j)$, that is,

$$f\left(y_t | \mathcal{I}_{t-1}\right) = \sum_{k=1}^{K} p_{t-1}\left(\Delta_t = j\right) f_N\left(y_t | \mathcal{I}_{t-1}, \Delta = k\right)$$

where $f_N(.)$ represents the density of a normal distribution. In order to facilitate the notation, consider that $h_t^{(k)}$ is the conditional variance when $\Delta_t = k$, that is,

$$h_t^{(k)} = \alpha_0^{(k)} + \alpha_1^{(k)} y_{t-1}^2 + \beta_1^{(k)} h_{t-1}^{(k)}. \tag{13}$$

Hence, the conditional variance of y_{t-1} given $\mathcal{I}_{t-2}$ is given by $h_{t-1} = \sum_{k=1}^{K} p_{t-2}(\Delta_{t-1} = k) h_{t-1}^{(k)}$. Gray (1996) replaced $h_{t-1}^{(k)}$ with h_{t-1} in Eq. (13). Thus, the specific conditional variance of each regime is given by:

$$h_t^{(k)} = \alpha_0^{(k)} + \alpha_1^{(k)} y_{t-1}^2 + \beta_1^{(k)} h_{t-1}. \tag{14}$$

and now h_{t-1} is not path dependent, so the likelihood estimation is tractable. Abramson and Cohen (2007) showed some inferential results of this model.

Haas et al. (2004) pointed out that the interpretation of the model proposed by Gray (1996) is problematic. As they mentioned, when one specifies a model with regime switching, the interest is to capture the dynamics in the conditional variance in periods of high and low volatility over time. Therefore, we are interested in the relationship of the parameters of the GARCH model corresponding to the variance of its regime $h_t^{(k)}$, similar to the GARCH model with a single regime described in Sect. 2. However, when using the conditional variance (14), $\beta_1^{(k)}$ cannot be interpreted as the persistence of $h_t^{(k)}$, since h_t is defined as the average of the conditional variances of all K regimes with respect to the probabilities of being in the k-th regime. Moreover, even if $\alpha_1^{(k)} = 0$, $h_t^{(k)}$ will be affected by shocks through changes in h_t, provided that $\beta \neq 0$.

In view of this, Haas et al. (2004) presented a new approach to MS-GARCH models with a clear interpretation in relation to the conditional variance of the regime and with feasible estimation.

3.1 Haas et al. (2004)'s model

Haas et al. (2004) proposed a new approach to univariate MS-GARCH models with a clear interpretation in relation to the conditional variance of the regime and feasible estimation. The MS(K)-GARCH(1,1) model proposed by Haas et al. (2004) can be expressed as:

$$y_t = \sqrt{h_t^{(\Delta_t)}} z_t, \tag{15}$$

$$h_t^{(k)} = \alpha_0^{(k)} + \alpha_1^{(k)} y_{t-1}^2 + \beta_1^{(k)} h_{t-1}^{(k)}, \tag{16}$$

where $h_t^{(k)}$ are the conditional variances of regime k. Note that in this definition, the term accompanying the parameter $\beta_1^{(k)}$, namely $h_{t-1}^{(k)}$, is what differentiates this model from the one proposed by Gray (1996). $\alpha_0^{(k)} > 0$, $\alpha_1^{(k)} \geq 0$, $\beta_1^{(k)} \geq 0$ for all $k = 1, \ldots, K$ are the sufficient conditions for $h_t^{(k)}$ to be positive; $\{z_t\}$ is a sequence of independent random variables with mean 0 and variance 1, commonly chosen as N(0.1). Finally, Δ_t is a Markov chain with finite state space $\mathcal{S} = \{1, \ldots, K\}$ and a transition matrix $K \times K$ P $:= [p_{ij}]$, where $p_{ij} = P(\Delta_t = j | \Delta_{t-1} = i)$, with each row summing to one and all transition probabilities strictly positive. Assuming that the Markov chain is irreducible and aperiodic, the stationary distribution $\boldsymbol{\pi}_\infty = (\pi_\infty^{(1)}, \pi_\infty^{(2)}, \ldots, \pi_\infty^{(K)})'$ can be obtained by $\boldsymbol{\pi}_\infty = (\boldsymbol{A}'\boldsymbol{A})^{-1}\boldsymbol{A}'\boldsymbol{v}$, where

$$\boldsymbol{A} = \begin{bmatrix} \boldsymbol{I}_k - \boldsymbol{P} \\ \mathbf{1}_k^{'} \end{bmatrix} \text{ and } \boldsymbol{v} = \begin{bmatrix} \mathbf{0}_k \\ \mathbf{1}_k \end{bmatrix}, \tag{17}$$

$\boldsymbol{I}_k$ is the $k \times k$ identity matrix, and $\mathbf{0}_k$ and $\mathbf{1}_k$ are $k \times 1$ vectors of zeros and ones, respectively.

In this model, the conditional variance $h_t^{(k)}$ depends only on the respective parameters of its regime *k*, in contrast to the one given by Gray (1996) in (14). Therefore, if $\max\{\beta_1^{(1)}, \ldots, \beta_1^{(K)}\} < 1$, like the GARCH models with a single regime, we have that $\alpha_1^{(k)}$ represents the impact of a shock at $t-1$ on $h_t^{(k)}$; $\beta_1^{(k)}$ represents the memory in $h_t^{(k)}$; and $\alpha_1^{(k)}(1-\beta_1^{(k)})^{-1}$ reflects the total impact of a unit shock on future variations of the regime *k*.

To study the second-order stationarity of the model, first consider the $K^2 \times K^2$ matrix $\boldsymbol{M} = [\boldsymbol{M}_{ij}]$, whose (i, j)-th block is given by $\boldsymbol{M}_{ij} = p_{ij}(\boldsymbol{\beta} + \boldsymbol{\alpha}_1 \boldsymbol{e}_k^{'})$, where $\boldsymbol{M}_{ij}$ is a $K \times K$ matrix, $\boldsymbol{\beta} = \text{diag}(\beta_1^{(1)}, \beta_1^{(2)}, \ldots, \beta_1^{(K)})$, $\boldsymbol{\alpha}_1 = (\alpha_1^{(1)}, \alpha_1^{(2)}, \ldots, \alpha_1^{(K)})'$, p_{ij} denotes the transition probability from state *i* to state *j*, and $\boldsymbol{e}_k$ is the $K \times 1$ unit vector with a one in its *k*-th position and zeros elsewhere. According to Haas et al. (2004), the process is weakly stationary if and only if $\rho(\boldsymbol{M}) < 1$, where $\rho(\cdot)$ denotes the spectral radius. Although the strict stationarity condition is complicated, Liu (2006) presented the conditions for strict stationarity. Haas et al. (2004) showed that a necessary condition for the weak stationarity of the MS(K)-GARCH(1,1) process is that $\beta_1^{(k)} < 1$ for all *k*, while Abramson and Cohen (2007) derived conditions for the strict and second-order stationarity of the MS(K)-GARCH(p,q) process. More information on the stationarity and properties of the model can be found in the Supplementary Material.

3.2 Non-robust estimators

In this section, we consider the maximum likelihood estimation of the model proposed by Haas et al. (2004). The maximum likelihood estimator is obtained by maximising the likelihood function

$$L(\boldsymbol{\theta}|\boldsymbol{y}_T) = \prod_{t=1}^{T} f(y_t \mid \mathcal{I}_{t-1}, \boldsymbol{\theta}) = \prod_{t=1}^{T} \sum_{k=1}^{K} [f(y_t \mid \Delta_t = k, \mathcal{I}_{t-1}, \boldsymbol{\theta}) P(\Delta_t = k|\mathcal{I}_{t-1})], \tag{18}$$

where $\boldsymbol{\theta} = (\boldsymbol{\alpha}_0, \boldsymbol{\alpha}_1, \boldsymbol{\beta}_1, p_{11}, \ldots, p_{KK})'$ is the parameter vector, and $f(.)$ denotes the conditional density of y_t. However, as the regime is unobserved, we need to infer the probability of being in time *t* in a given regime $P(\Delta_t = k \mid \mathcal{I}_{t-1})$. This is accomplished using the filter of Hamilton (1989), a nonlinear recursive filter for estimating the state probabilities in a Markov-switching autoregressive model.

3.2.1 Estimation via Hamilton filter

Define $\boldsymbol{\xi}_t$ as the vector of conditional probabilities of being in each regime at time t, that is $\boldsymbol{\xi}_t = (P(\Delta_t = 1|\mathcal{I}_{t-1}), \ldots, P(\Delta_t = K \mid \mathcal{I}_{t-1}))'$ and $\boldsymbol{h}_t$ as the vector of the conditional variances of each regime, that is, $\boldsymbol{h}_t = (h_t^{(1)}, \ldots, h_t^{(K)})'$. Thus, given the initial values $(y_0, \boldsymbol{h}_0, \boldsymbol{\xi_0})'$ (or $(y_1, \boldsymbol{h}_1, \boldsymbol{\xi_1})'$), the Hamilton filter estimates $\boldsymbol{\xi}_t$ recursively as follows:

(1) Given the vector of conditional probabilities $\boldsymbol{\xi}_{t-1} = (\xi_{1,t-1}, ..., \xi_{K,t-1})'$:

$$\xi_{i,t-1} = P\left(\Delta_{t-1} = i|\mathcal{I}_{t-1}\right). \tag{19}$$

(2) Calculate the conditional variances of each regime:

$$h_t^{(k)} = \alpha_0^{(k)} + \alpha_1^{(k)} y_{t-1}^2 + \beta_1^{(k)} h_{t-1}^{(k)}. \tag{20}$$

Under the normality assumption, calculate the conditional densities of each regime.

$$\eta_{it} = f\left(y_t|\Delta_t = i, \mathcal{I}_{t-1}\right) = \left(2\pi h_t^{(i)}\right)^{-1/2} \exp\left\{-y_t^2/(2h_t^{(i)})\right\}. \tag{21}$$

The normality assumption is not the only option. For instance, Paolella et al. (2012) used the Student-t distribution. If the innovations are not normally distributed but instead exhibit excess kurtosis, assuming normality within each regime can seriously affect the identification of the underlying regime and, consequently, the estimation of the regime-specific parameters. That is, a model assuming normally distributed regime-specific innovations will tend to detect regime changes excessively often in response to atypical observations (Paolella et al. 2012).

(3) Calculate the conditional density of the t-th observation as:

$$f\left(y_t|\mathcal{I}_{t-1}\right) = \sum_{i=1}^{k}\sum_{j=1}^{k} p_{ij} P\left(\Delta_{t-1} = i|\mathcal{I}_{t-1}\right)\eta_{jt} = \mathbf{1}'\left(\boldsymbol{P}\boldsymbol{\xi}_{t-1} \odot \boldsymbol{\eta}_t\right), \tag{22}$$

where $\odot$ represents the Hadamard product and $\boldsymbol{P}$ is the transition matrix.

(4) Update the probabilities of being in each regime as:

$$\boldsymbol{\xi}_t = \frac{\boldsymbol{P}\boldsymbol{\xi}_{t-1} \odot \boldsymbol{\eta}_t}{\mathbf{1}'\left(\boldsymbol{P}\boldsymbol{\xi}_{t-1} \odot \boldsymbol{\eta}_t\right)}. \tag{23}$$

By iterating through steps (1)–(4), the conditional quasi-likelihood function can be computed over the entire sample. The estimate of $\boldsymbol{\theta}$ is then obtained by maximising the likelihood or log-likelihood function using a numerical optimisation method. In such a way that:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\Theta} \sum_{t=2}^{T} \log f\left(y_t|\mathcal{I}_{t-1}\right) = \arg\max_{\boldsymbol{\theta}\in\Theta} \sum_{t=2}^{T} \log\left(\mathbf{1}'\left(\boldsymbol{P}\boldsymbol{\xi}_{t-1}\odot\boldsymbol{\eta}_{jt}\right)\right). \quad (24)$$

As with single-regime models, the choice of initial values can influence parameter estimates. For the initial values of $\boldsymbol{\xi}_1$, the estimated stationary distribution of the Markov chain is often used (17), or else considered to be equiprobable, or considered to be parameters and estimated by QML. For the values of $\boldsymbol{h}_1$, we can assume that they are a function of the parameters, such that for the regime k we have that $h_1^k = \alpha_{0k}/(1-\alpha_{1k}-\beta_{1k})$.

3.2.2 Notes on quasi-maximum likelihood estimation

Krabbe (2025) established the consistency and asymptotic normality of the QML estimator for a general class of Markov-switching observation-driven models under suitable regularity conditions. The MS-GARCH model proposed by Haas et al. (2004) is a special case of this framework, implying that its QML estimator inherits these asymptotic properties.

Paolella et al. (2012) pointed out that, as in the GARCH mixing models, log-likelihood maximisation (24) is often performed by an optimisation method based on the Hessian matrix with numerically computed derivatives. Because the likelihood function may have several plausible local maxima, it is necessary to use multiple sets of initial values to increase the likelihood of identifying the global maximum. For a slightly different model, Augustyniak (2014) also highlighted the impact of initial values on parameter estimates and proposed an algorithm to reduce sensitivity to initial values. Another point is that, due to the nature of the likelihood function as a mixture, it may have singularities (infinite likelihood values).

Andreou and Ghysels (2009) presented several statistical methods for determining the number of regimes. However, Paolella et al. (2012) argued that these standard tests, as well as likelihood ratio tests, do not perform well, whereas information criteria such as AIC and BIC perform reasonably well in this context. On the other hand, in empirical applications, the number of regimes is usually chosen to be two for reasons of interpretability. When $K > 2$, at least one of the regimes tends to have a very small stationary probability, making its parameter estimates subject to large estimation errors (Paolella et al. 2012).

3.3 Robust estimation and prediction

We propose a robust estimator that incorporates three modifications to the estimator presented previously: (i) the BIP method given by Eq. (6) in Sect. 2.2 to minimise the effect of outliers on the conditional variance estimates for each regime; (ii) the use of the Student-t distribution with four degrees of freedom for the conditional densities; and (iii) the objective function proposed by Boudt et al. (2013) for the univariate case.

The motivation for using the BIP is the same as in the single-regime case. The use of the Student-t distribution with four degrees of freedom in the conditional densities

is motivated by robust filtering techniques. Several authors have pointed out that conventional filters, including Hamilton's filter, are not robust and may perform poorly even under relatively mild contamination. For example, Petrus (1999), Zou et al. (2000), and Calvet et al. (2015) demonstrated the poor performance of several filtering methods, including least-squares and Bayesian filters, in the presence of outliers.

To understand the representation of outliers in filters, we follow Calvet et al. (2015). Let $y_t \in \mathbb{R}$ denote the observation with conditional density $f(y_t|\Delta_t, \mathcal{I}_{t-1})$. As in (23), Bayes' rule implies that the probabilities of being in each regime inferred by the filter satisfy:

$$\lambda\left(\Delta_t|y_t, \mathcal{I}_{t-1}\right) = \frac{f\left(y_t|\Delta_t, \mathcal{I}_{t-1}\right)\lambda\left(\Delta_t|\mathcal{I}_{t-1}\right)}{f\left(y_t|\mathcal{I}_{t-1}\right)}, \tag{25}$$

where $f(y_t|\mathcal{I}_{t-1}) = \int f(y_t|\Delta_t, \mathcal{I}_{t-1})\lambda(\Delta_t|\mathcal{I}_{t-1})d\Delta_t$. Thus, let $f_{cont}(.|\Delta_t, \mathcal{I}_{t-1}, \omega)$ denote a family of observation densities contaminated by outliers, parameterised by ω, where ω belongs to an interval $\mathcal{D}$ on the real line containing zero. If $\omega = 0$, the contaminated density coincides with the uncontaminated observation density, that is, $f_{cont}(y_t|\Delta_t, \mathcal{I}_{t-1}, \omega = 0) = f(y_t|\Delta_t, \mathcal{I}_{t-1})$ for all Δ_t and $\mathcal{I}_{t-1}$. When contamination is not taken into account, the naive application of Bayes' rule leads to formula (25).

To robustify the filter, Calvet et al. (2015) and Petrus (1999) employed robust densities. Calvet et al. (2015) used a specification based on limiting the derivative of the log-density, while Petrus (1999) adopted one based on Huber's objective function, $\rho(u) = u^2 I(|u| \leq k) + k|u|I(|u| < k)$. Analogously to Boudt et al. (2013), we use a Student-t distribution with four degrees of freedom.

Therefore, the proposed robust estimation procedure modifies the estimation method presented in Sect. 3.2.1 in three respects. First, in Step 2, the BIP mechanism is used to minimise the effect of outliers on the conditional variance estimates for each regime. Second, the Student-t distribution with four degrees of freedom is used to compute the conditional density of each regime. Finally, instead of maximising the log-likelihood function in (24), we minimise a modified objective function that combines the log-likelihood with the function ρ defined in (8). Accordingly, the estimate of $\boldsymbol{\theta}$ is obtained by minimisation, that is,

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}\in\Theta}{\arg\min} \sum_{t=1}^{T} \log\left[\sum_{i=1}^{k}\sum_{j=1}^{k} p_{ij} P\left(\Delta_{t-1} = i|\mathcal{I}_{t-1}\right) \frac{1}{\sqrt{h_t^{(j)}}}\left(1 + \frac{y_t^2}{2h_t^{(j)}}\right)^{-\frac{(1+4)\sigma_4}{2}}\right]. \tag{26}$$

where σ_4 is given by (10).

To estimate the one-step-ahead volatility, we propose using the robust BIP mechanism given by Eq. (6). From the second step onwards, we can use the procedure of Paolella et al. (2012), which is described in the Supplementary Material, because from the second-step we use the expected values, which are not affected by outliers. Using the one-step-ahead conditional distribution of each regime, we can use numerical methods or path simulation to estimate the one-step-ahead VaR. For forecasts of two or more steps ahead, path simulation is required.

4 Simulation

In this section, we investigate the performance of the estimators for the MS-GARCH models in the presence and absence of additive outliers. We use GARCH(1,1) models for both regimes. As noted by Francq and Zakoian (2019, Section 8.5, p. 235), the GARCH(1,1) model is chosen because it is by far the most widely used model by practitioners for estimating the volatility of daily returns. This choice is motivated by the widely held view that the GARCH(1,1) specification is sufficient to capture the main characteristics of financial time series, whereas higher-order models may introduce unnecessary complexity.

4.1 Data generator process

We consider a sample size of $T = 3000$ and 1000 Monte Carlo replications. Given the choices $p_{11} = 0.96$ and $p_{22} = 0.98$, the stationary distribution of the Markov chain is $\boldsymbol{\pi}_\infty = (1/3, 2/3)^{'}$, so that the expected number of observations in each regime is 1000 and 2000, respectively.The observed series y_t^*, with additive outliers, is defined as:

$$y_t^* = y_t + \text{sign}(y_t) I_A(t)\, d_t \sqrt{h_t^{(\Delta_t)}},$$

where y_t is given by (15), $\text{sign}(.)$ represents the sign function, $I_A(t)$ is an indicator variable representing the position of the outliers, with A being the set of observations affected by outliers, d_t is a constant belonging to $\mathbb{R}_+$ representing the magnitude of the outliers in terms of conditional volatility, and $\sqrt{h_t^{(\Delta_t)}}$ is defined by (16).

Some authors consider isolated and consecutive outliers placed at strategic locations, such as the beginning, middle, and end of the series, to assess their respective effects. As in Muler and Yohai (2008) and Boudt et al. (2013), we evaluate the estimators when a fraction of observations is contaminated. The fraction, say ϵ, is set to $\epsilon = 1\%$, $\epsilon = 5\%$, $\epsilon = 10\%$ and $\epsilon = 0\%$ in the absence of outliers. The positions of the outliers are sampled uniformly at random without replacement. Finally, we consider two values of d, 3 and 5. The data generating process is:

$$y_t = \sqrt{h_t^{(\Delta_t)}} z_t \tag{27}$$

$$h_t^{(1)} = 2.0 + 0.10\, y_{t-1}^2 + 0.6\, h_{t-1}^{(1)} \tag{28}$$

$$h_t^{(2)} = 0.3 + 0.35\, y_{t-1}^2 + 0.2\, h_{t-1}^{(2)}. \tag{29}$$

We consider two innovation distributions: the standard normal distribution and the standardised Student-t distribution with five degrees of freedom. However, for the sake of conciseness, only a subset of the results is reported for each distribution. Although the models are not identical, the simulations use the same parameter values

as those employed in the data-generating processes considered by Bauwens et al. (2010), Augustyniak (2014), and Billio et al. (2016).

The unconditional variance of the first regime is 10 times greater than that of the second. That is, we have $\alpha_0^{(1)}/(1-\alpha_1^{(1)}-\beta_1^{(1)}) = 20/3$ and $\alpha_0^{(2)}/(1-\alpha_1^{(2)}-\beta_1^{(2)}) = 2/3$.

4.2 Estimators

A robust estimator (i) should have reasonably good efficiency (optimal or nearly optimal) when the assumed model is correct; (ii) should be robust, in the sense that small deviations from the model's assumptions only slightly affect the estimates; and (iii) should not exhibit catastrophic performance under moderately larger deviations from the model. We consider three estimators in the simulation: QML-n and QML-t estimators defined in Sect. 3.2, and the robust (Rob) estimator defined in Sect. 3.3. A modification was introduced in the function $r_{k_{\delta,1}}(u)$, defined in Eq. (6), so that the estimators produce better results in estimating the parameters. In the literature, some authors propose replacing the value of $k_{\delta,1}$ when $x > k_{\delta,1}$, like Muler and Yohai (2008) and Boudt et al. (2013), for example. However, others suggest using 1 (expected value) when $x > k_{\delta,1}$, such as Carnero et al. (2012). This value corresponds to approximately $k_{0.69,1}$. In the simulation, we use $\delta = 0.82$, which lies between 0.69 and 0.95. In the empirical application (Sect. 5), we also consider $\delta = 0.90$ and 0.975, and provide guidance on how to select an appropriate value depending on the objective of the application.

4.3 Initial values

As noted by Paolella et al. (2012), there may be several "plausible" local maxima of the MS-GARCH model's likelihood function. Thus, in each replication of the simulation study used to evaluate the performance of the estimators, we consider several different sets of initial values and evaluate the likelihood function for each of them. For the three initial values that yield the highest likelihood (if it is a loss function, the lowest), we estimate the parameters giving us three parameter estimates generated by three different initial values. Among these three values, we use the following procedure to determine which estimate corresponds to the global maximum. The selection procedure is based on the following principle. If the three estimation runs converge to the same solution, that is, they produce identical parameter estimates and the same final likelihood value, the corresponding estimate is retained. Otherwise, if the three runs converge to different solutions, we retain the estimate associated with the highest final likelihood value. If there is no difference between the final likelihood values, we use the average of the three parameter estimates.

Here, we consider the parameters and the final likelihood to be different if there is a difference greater than 10^{-4} either in the parameters or in the final likelihood, except for the intercept of the high volatility regime, which will be different if there is a difference greater than 10^{-2}.

The cases where the three estimates do not converge to the same place are exceptions. For instance, we generated 1000 Monte Carlo replications of the process defined in Sect. 4.1 and applied the above mechanism using maximum likelihood estimation. In only 27 replications (2.7%), the estimation runs did not converge to the same solution. Of these, only 14 (1.4% of all replications) converged to the same likelihood value but yielded different parameter estimates.

The performance of the Hamilton's filter is analysed in Sect. 4.4, whereas the accuracy of the parameter estimates is examined in Sect. 4.5.

4.4 Analysis of the filter performance

Figure 1 presents the proportion of correct classifications for the data-generating process described in Sect. 4.1 with Gaussian distributions, considering (a) only the observations affected by outliers and (b) all observations. We set a threshold of 0.50 to classify whether an observation is in one regime or not. As expected, the filters perform better in the high-volatility regime, and in this regime, the performance tends to improve with greater contamination and larger outlier sizes since the effect of outliers is to classify the observation as generated by the high-volatility regime.

For the low-volatility regime with 1% contamination, the QML-n estimator achieves less than 14% correct classifications, even when the outlier magnitude is $d = 3$. In contrast, the QML-t and Robust estimators attain hit rates above 75% for $d = 3$ and above 48% for $d = 5$. When the degree of contamination increases to 5%, the performance of all estimators deteriorates. The QML-t and robust estimators still perform reasonably well for $d = 3$, achieving more than 60% correct classifications, but this percentage drops to less than 17% when $d = 5$. Finally, when the proportion of outliers reaches 10% and $d = 3$, the robust estimator performs similarly to the 5% contamination case and outperforms QML-t. However, when $d = 5$, all estimators perform poorly.

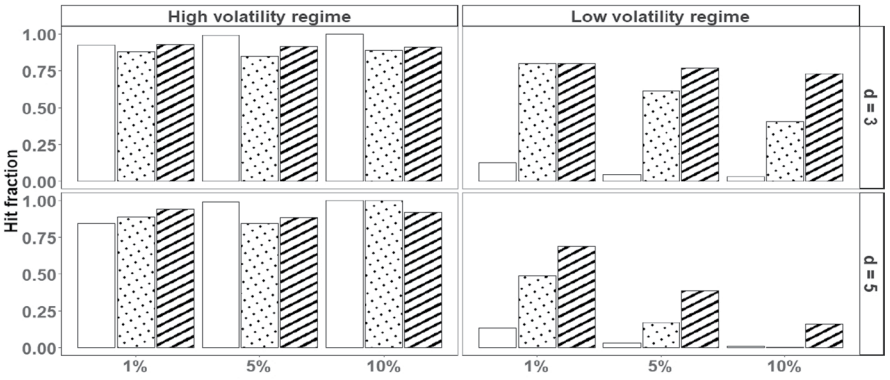
In the high-volatility regime, all three estimators generally perform well, with similar performance.

When all observations are considered, all estimators perform well in the low-volatility regime. However, the robust estimator generally performs slightly better than the two QML estimators. In the high-volatility regime, the QML-t and robust estimators outperform the QML-n estimator, except when the proportion of outliers is 10% and d=5.

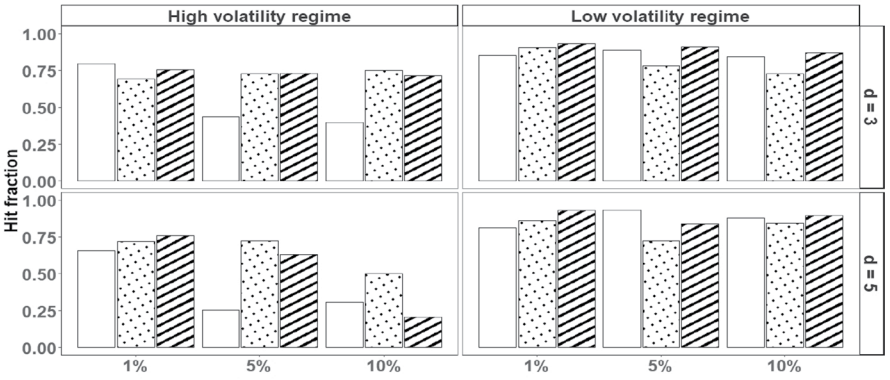
4.5 Models estimation

The bias and RMSE of the parameter estimates obtained from the Monte Carlo simulation are reported in the Supplementary Material. Figures 2 and 3 present boxplots of the parameter estimates for the MS-GARCH model with Gaussian innovations in the high- and low-volatility regimes, respectively. The corresponding results for the standardised Student-t distribution are presented in Figs. 4 and 5.

When there are no outliers, as expected, QML-n and QML-t present the best performance when the innovations follow Gaussian and Student-t distributions, respectively. However, when the innovations follow a Student-t distribution, the per-



(a) contaminated observations

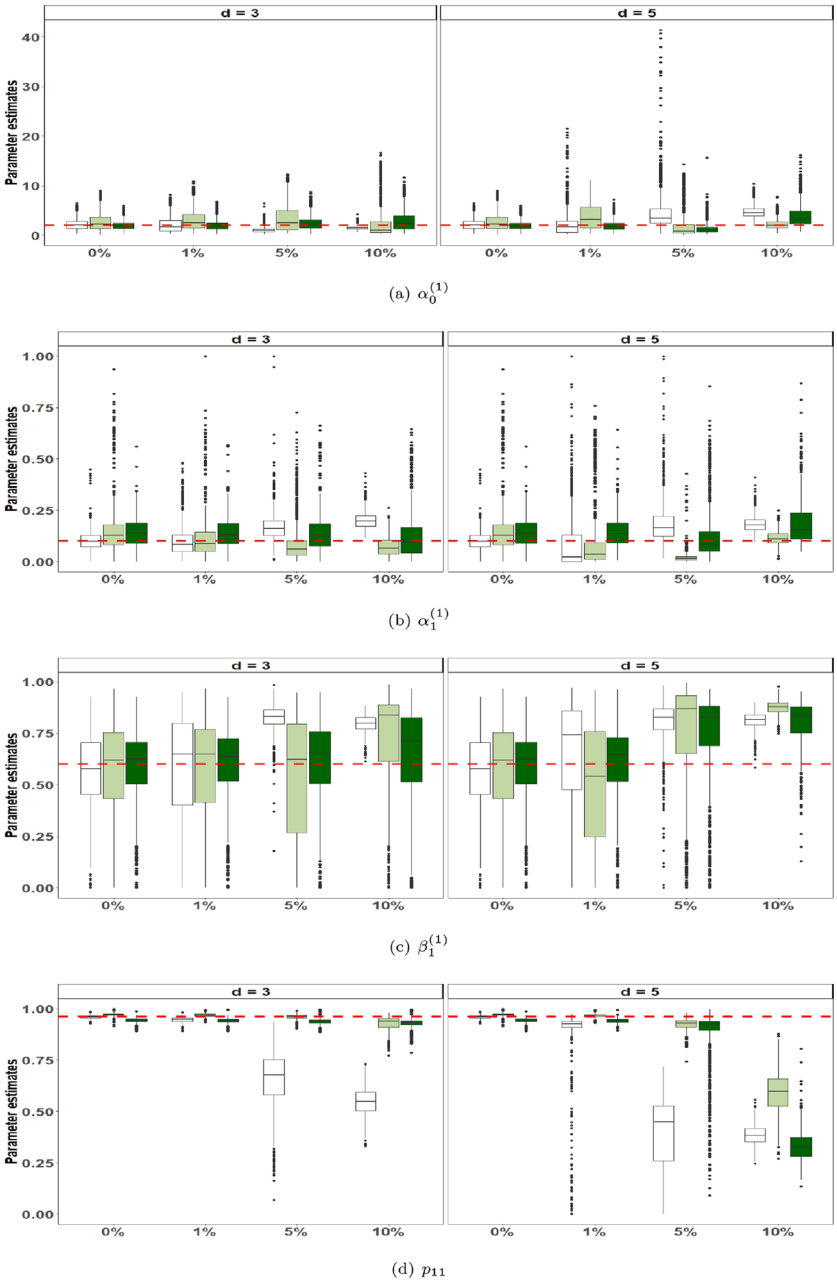


(b) all observations

Fig. 1 Proportion of correct classification of the different estimators for the MS-GARCH model in the **a** contaminated observations **b** all observations. The horizontal axis gives the percentage of outliers. The empty bars denote the QML-n estimator, the bars with dots represent the QML-t estimator and those with diagonal lines denote the robust estimator. The graphs on the left indicate when the correct regime has high volatility, that is, regime 1, and those on the right indicate regime of low volatility, regime 2. The upper graphs show outliers of size 3 times the conditional standard deviation, while the lower ones show 5 times the conditional standard deviation

formance of QML-n is poor for both regimes. In contrast, when the innovations follow a Gaussian distribution, the performance of QML-t is reasonable for estimating the high-volatility regime but poor for the low-volatility regime. The robust estimator does not perform as well as QML-n and QML-t when the innovation distribution is correctly specified, but it is less affected when the distribution is misspecified.

Under contamination, the robust estimator becomes increasingly competitive, particularly at higher contamination levels. The performance of QML-n and QML-t deteriorates, but the degree of deterioration depends on the regime, the innovation distribution, the percentage of outliers, and their sizes. As expected, the performance is worst when the distribution of the innovations is incorrectly specified and when the percentage and size of outliers increase.



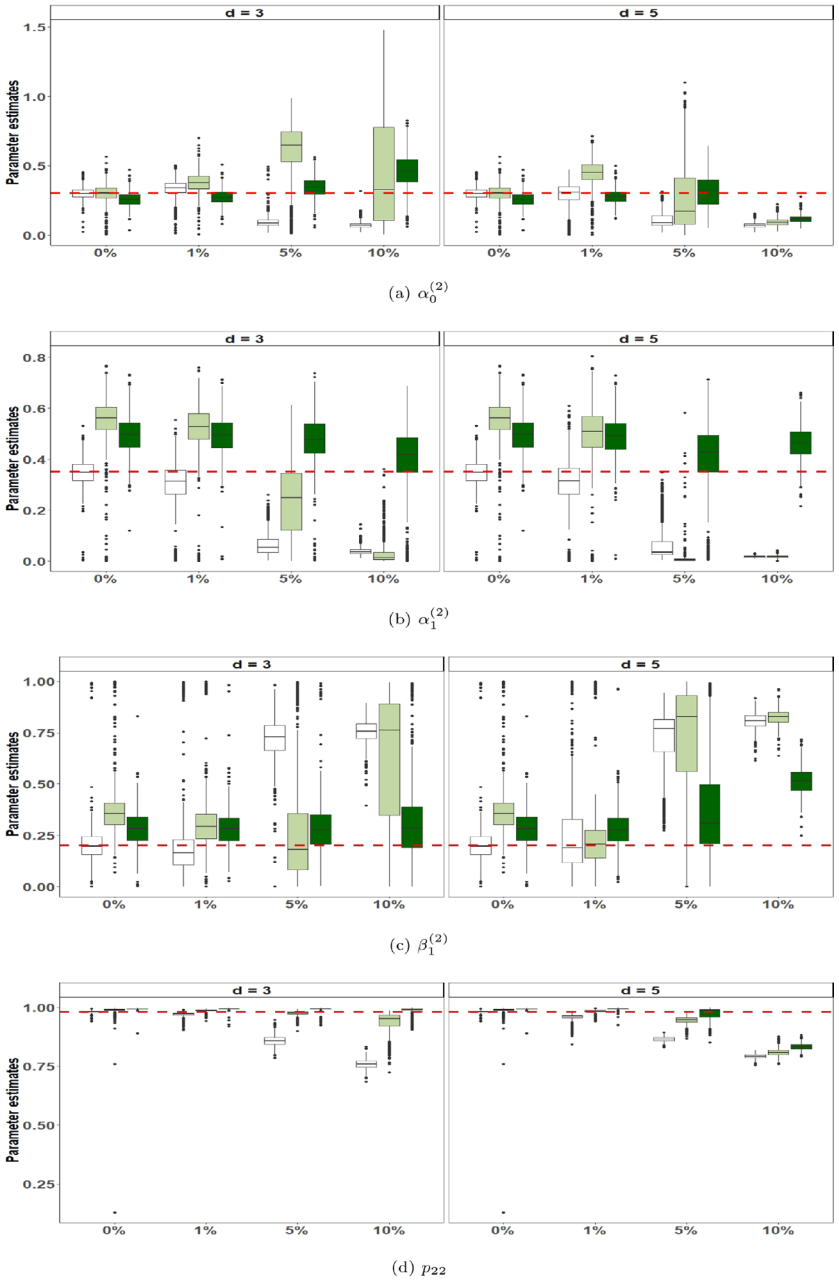
(a) $\alpha_0^{(1)}$

(b) $\alpha_1^{(1)}$

(c) $\beta_1^{(1)}$

(d) p_{11}

Fig. 2 Boxplot of parameter estimates for the high-volatility regime from the Monte Carlo replications of the MS-GARCH(1,1) model with Gaussian innovations. The x-axis represents the percentage of outliers, and the y-axis represents the parameter estimate values. The left panel shows results for outliers of size $d = 3$, and the right panel shows results for $d = 5$. The QML-n estimator is shown in white, the QML-t estimator in light green, and the robust estimator in dark green. The red dashed line represents the true parameter value. The parameters in each panel are: $\mathbf{a}\alpha_0^{(2)}$, $\mathbf{b}\alpha_1^{(2)}$, (c) $\beta_1^{(2)}$, and $\mathbf{d}p_{22}$.



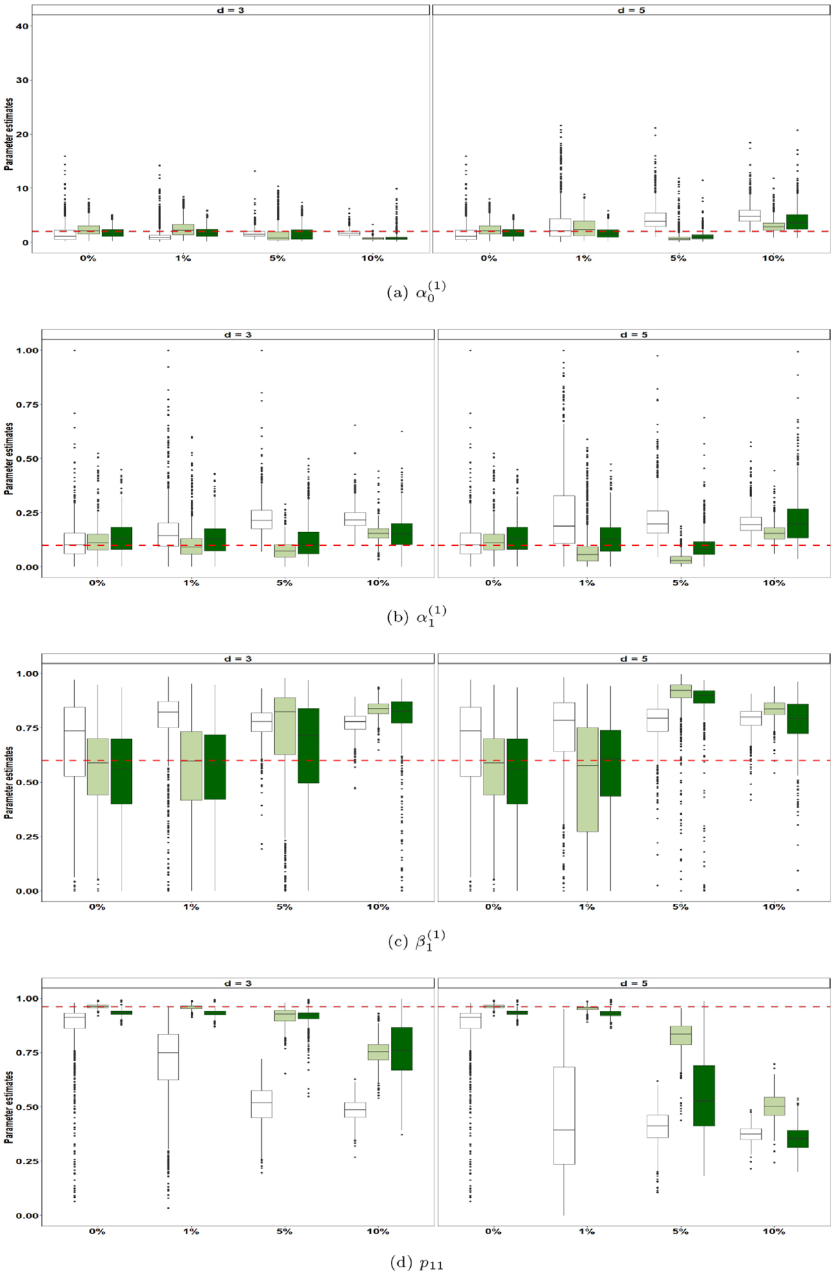
(a) $\alpha_0^{(2)}$

(b) $\alpha_1^{(2)}$

(c) $\beta_1^{(2)}$

(d) p_{22}

Fig. 3 Boxplot of parameter estimates for the low-volatility regime from the Monte Carlo replications of the MS-GARCH(1,1) model with Gaussian innovations. The x-axis represents the percentage of outliers, and the y-axis represents the parameter estimate values. The left panel shows results for outliers of size $d = 3$, and the right panel shows results for $d = 5$. The QML-n estimator is shown in white, the QML-t estimator in light green, and the robust estimator in dark green. The red dashed line represents the true parameter value. The parameters in each panel are: **a**$\alpha_0^{(2)}$, **b**$\alpha_1^{(2)}$, **c**$\beta_1^{(2)}$, and **d**p_{22}



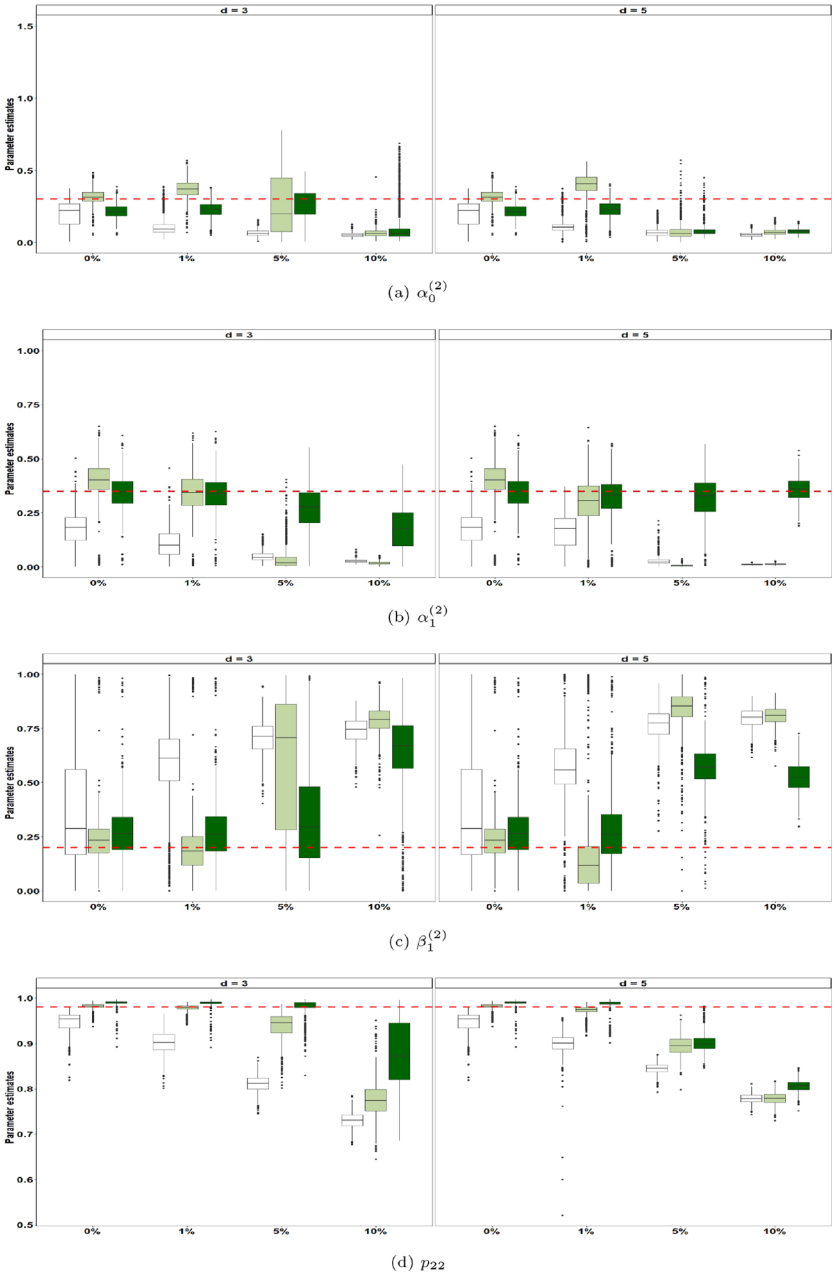
(a) $\alpha_0^{(1)}$

(b) $\alpha_1^{(1)}$

(c) $\beta_1^{(1)}$

(d) p_{11}

Fig. 4 Boxplot of parameter estimates for the high-volatility regime from the Monte Carlo replications of the MS-GARCH(1,1) model with Student-t innovations. The x-axis represents the percentage of outliers, and the y-axis represents the parameter estimate values. The left panel shows results for outliers of size $d = 3$, and the right panel shows results for $d = 5$. The QML-n estimator is shown in white, the QML-t estimator in light green, and the robust estimator in dark green. The red dashed line represents the true parameter value. The parameters in each panel are: **a**$\alpha_0^{(2)}$, **b**$\alpha_1^{(2)}$, **c**$\beta_1^{(2)}$, and **d**p_{22}



(a) $\alpha_0^{(2)}$

(b) $\alpha_1^{(2)}$

(c) $\beta_1^{(2)}$

(d) p_{22}

Fig. 5 Boxplot of parameter estimates for the low-volatility regime from the Monte Carlo replications of the MS-GARCH(1,1) model with Student-t innovations. The x-axis represents the percentage of outliers, and the y-axis represents the parameter estimate values. The left panel shows results for outliers of size $d = 3$, and the right panel shows results for $d = 5$. The QML-n estimator is shown in white, the QML-t estimator in light green, and the robust estimator in dark green. The red dashed line represents the true parameter value. The parameters in each panel are: $\mathbf{a}\alpha_0^{(2)}$, $\mathbf{b}\alpha_1^{(2)}$, $\mathbf{c}\beta_1^{(2)}$, and $\mathbf{d}p_{22}$

In the case of the Gaussian distribution, even with only 1% contamination, no method consistently outperforms the others, and the QML-n estimator is no longer the best performer. In the case of the Student-t distribution, the performance of the QML-n estimator deteriorates further, while the QML-t and robust estimators show comparable performance, with neither consistently dominating the other across all scenarios.

As the proportion of contamination increases, the performance of the QML-n estimator deteriorates substantially; therefore, we no longer discuss this estimator. With 1% contamination, the robust estimator outperforms the QML-t estimator. The QML-t estimator performs particularly poorly in the low-volatility regime.

With 10% contamination, none of the estimators performs well. However, the robust estimator still performs substantially better than the other estimators.

The literature includes several Monte Carlo simulation studies on the estimation of MS-GARCH model parameters. Augustyniak (2014) presented simulation results using the specification proposed by Gray (1996), while Billio et al. (2016) reported results based on Bayesian estimation methods. In both cases, there was difficulty in estimating the parameters $\alpha_0^{(1)}$, $\beta_1^{(1)}$ and $\beta_1^{(2)}$, even in the absence of *outliers*. In our simulations, we also observed that, in general, these parameters were the most problematic. Additionally, the BIP mechanism tended to perform better when the proportion of outliers was higher. Furthermore, the QML-t estimator performed well in estimating the Markov chain transition matrix probabilities.

Estimating MS-GARCH models is known to present several difficulties. Four main reasons can be highlighted: (i) estimating nonlinear GARCH models requires substantially more data than, for instance, linear ARMA models; (ii) only a fraction of the observations belongs to each regime; (iii) classification errors; and (iv) observations from each regime appear in clusters rather than as consecutive values.

To illustrate these difficulties, consider the estimation of the parameter $\alpha_0^{(1)} = 2.0$ in Augustyniak (2014) and Billio et al. (2016). Although they used different Markov-switching GARCH specifications and a smaller sample size of 1500, the volatility models within each regime were the same. For a single series, Billio et al. (2016) reported a mean and standard deviation of the Bayesian posterior distribution equal to 4.240 and 0.960, respectively, using the particle MCMC method (see the third column of their Table 2). In Augustyniak (2014), based on 200 replications, the mean estimate was 3.733 with an MSE of 1.987 (see their Table 3). Similar difficulties were reported for other parameters. All these results were obtained from simulated data without outliers.

4.6 Computational cost

To illustrate the computational cost, for the MS-GARCH(1,1) model with Gaussian innovations, we compare the execution times of three estimation methods: the robust estimator, the QML that incorporates an initialisation procedure for the starting values (QML-IP), as defined in Sect. 4.3 and the standard QML estimator. Table 1 reports summary statistics for the execution time across different sample sizes.

The results indicate that QML presents the lowest computational cost across all scenarios, exhibiting the smallest mean and median execution times. For $T = 1500$, the median execution time of the QML estimator (0.043) is substantially lower than those of the QML-IP estimator (0.234) and the robust estimator (0.613). This pattern remains consistent as the sample size increases.

Overall, the results highlight differences in the computational requirements of the three estimation methods. The standard QML estimator has the lowest execution times, whereas the robust estimator and the QML-IP estimator are more computationally demanding owing to their more complex estimation procedures. Nevertheless, although the relative differences in execution time are substantial, the absolute computational cost of all three methods remains acceptable.

5 Application

The focus of the application is to compare the performance of the MS-GARCH model forecasts estimated by QML, presented in Sect. 3.2, with those from the robust version presented in Sect. 3.3, in estimating VaR. We also use the application to discuss the calibration of the parameter δ.

To this end, we consider three values for the parameter δ: 0.820, used in the simulation study; 0.900, as suggested by Muler and Yohai (2008) and Boudt et al. (2013); and 0.975, also suggested by Boudt et al. (2013). Including the QML-n and QML-t estimators, this results in a total of five estimators.

We compare the performance of the estimators using four different exchange rate series, namely the Euro, Yen, Swiss Franc, and Pound series, all quoted against the US dollar at noon in New York.

For all series, data span the period from 01/04/1999 to 05/28/2021 and were obtained from the website https://www.federalreserve.gov/releases/H10/hist/. The bilateral exchange rate data are updated every Monday, yielding a total of 5,845 observations. This exercise is mainly based on Haas et al. (2004) and Paolella et al.

Table 1 Summary statistics of execution times (in seconds) for the robust estimator, the standard QML estimator, and the QML with initialization procedure (QML-IP), as defined in Sect. 4.3, across different sample sizes

T	Estimador	Mean	Median	Standard Deviation
1500	Rob	0.664	0.613	0.145
	QML-IP	0.269	0.234	0.081
	QML	0.049	0.043	0.019
3000	Rob	1.157	1.089	0.213
	QML-IP	0.405	0.358	0.109
	QML	0.084	0.073	0.031
5000	Rob	1.849	1.746	0.332
	QML-IP	0.601	0.533	0.163
	QML	0.138	0.121	0.045

Results are based on Monte Carlo simulations and reported in terms of mean, median, and standard deviation. MS-GARCH(1,1) model with Gaussian innovations

(2012). Paolella et al. (2012) used moving windows of size 2500, while Haas et al. (2004) used expanding windows starting with a size of 3000.

During the period, several crises can be identified: the attacks on the World Trade Center and the Pentagon in 2001; the period of economic uncertainty in Japan in the 1990 s, marked by rising unemployment rates, bank failures, and related issues; the 2008–2009 subprime crisis, which caused a shock to the global economy; and the COVID-19 pandemic after 2020.

In the analysis, we use log-returns expressed in percentage terms. Except for the euro, all series exhibit kurtosis greater than that of the normal distribution (> 3), with the Swiss franc and the pound showing the highest values. All series exhibit low skewness, and although the sample variances are approximately similar across the four series, the Swiss franc series shows a much higher kurtosis (33.01), followed by the pound series (10.79). Some descriptive statistics are presented in the Supplementary Materials.

Figures showing the daily evolution of the exchange rates, their log-returns, the sample autocorrelation function of the log-returns, and the sample autocorrelation function of the squared log-returns for the euro, pound, yen, and Swiss franc are also provided in the Supplementary Materials. In the time-series plots, one can observe volatility clustering, as well as differences in the level of each series. We highlight the yen series, which was already declining due to the effects of the subprime crisis before being further affected in March 2011 by the earthquake and the resulting tsunami in Japan.Later, in 2013, the government announced an economic stimulus plan aimed at helping the country recover from the crisis and restoring the currency to its pre-crisis level. One can also observe the impact of the 2008–2009 subprime crisis on these series, with this period exhibiting higher volatility across all four series. For the Swiss franc log-return series, there is a negative return that stands out from the others. This return corresponds to 15 January 2015, when the Swiss National Bank (SNB) announced the abandonment of its policy of maintaining a ceiling on the exchange rate of the Swiss franc against the euro.

The first-lag autocorrelations of the returns for all four currencies are statistically significant. Regarding the squared returns, the autocorrelations are also statistically significant and exhibit a high degree of persistence, particularly for the euro series.

Because we work with MS-GARCH models without a mean specification, we first filtered the conditional mean of the series. The filter used for all series was an AR(1). Let r_t denote the return at time t. The filtered return, r_t^*, is then given by $r_t^* = r_t - (\phi r_{t-1})$, where ϕ is the autoregressive parameter, estimated by maximum likelihood. Note that we use an AR(1) model without a constant term. An analysis (not shown) of the autocorrelation functions of r_t^* indicates that this filter is sufficient to remove serial correlation.

The forecasting performance of these models is evaluated using the Value-at-Risk (VaR) measure. To compute the VaR, we follow the approach of Ardia et al. (2018) for MS-GARCH models. Under these models, the one-step-ahead conditional probability density function of y_{T+1} is a mixture of K regime-dependent distributions, that is,

$$f\left(y_{T+1}|\mathcal{I}_T\right) = \sum_{k=1}^{K} \pi_{k,T+1} f\left(y_{T+1}|\Delta = k, \mathcal{I}_T\right),$$

where $\pi_{k,T+1}$, the mixing weight, is defined as $\pi_{k,T+1} \equiv \sum_{k=1}^{K} p_{i,k}\eta_{i,T}$, such that $p_{i,j}$ denotes the transition probability from state i to state j, and $\eta_{i,T}$ is the filtered probability $P[\Delta_T = i|\mathcal{I}_{T-1}]$ at time *t*.

We are interested in evaluating the forecasting performance of the model for one-step-ahead predictions for each series, using VaR levels $\alpha = 0.01$ and $\alpha = 0.05$. For this purpose, each model is estimated using a moving window, from which the one-step-ahead VaR is obtained. Using estimation windows of 3,000 observations, we obtain a total of 2,884 VaR forecasts in the backtesting exercise. Initially, for each window, missing values are imputed using a level-and-trend model based on the smoothed estimates provided by the Kalman filter, and an AR(1) filter is then applied to ensure a zero conditional mean (yielding demeaned series). We denote the last observation of the estimation window by *T*. If the observation at time *T* and/or $T + 1$ is missing, we do not compute the one-step-ahead VaR forecast. Otherwise, we estimate the model using the window of observations up to time *T* to evaluate the VaR.

We present the results of the MS(2)-GARCH(1,1) model for the robust case and for estimation using the Gaussian and Student-t QML estimators. The estimated model parameters for each of the four currencies in the first and last windows are reported in the Supplementary Material. The aim is to illustrate how the parameters vary over time. This variation supports our use of moving windows. There are large changes in the parameter estimates from the first to the last window for all estimators. Furthermore, the QML-n and robust estimators tend to assign a much higher stationary probability to one regime than to the other. For the regime with higher probability, the estimates are similar across methods.

5.1 Estimation of the probability of the regimes

To illustrate the estimated probabilities of being in each regime for the first window of the yen series, Fig. 6 shows, for simplicity, only the probabilities of being in the high-volatility regime for $\delta = 0.820$. The QML-n estimator assigns a low probability of being in the high-volatility regime for almost all observations. However, for a relatively large number of isolated observations, the estimated probabilities are close to one. Since this behaviour is not expected in financial markets, it suggests that the QML-n estimator is unstable. The estimates from the other two estimators are more stable, indicating a high probability of being in the low-volatility regime throughout the entire period. However, they also exhibit different behaviours: while the QML-t estimates remain within a narrower range over time, the robust estimator displays greater local stability.

5.2 Analysis of VaR estimates and forecasts

For the VaR analysis, we use log-returns filtered by an AR(1) model. That is, for the price P_t and its log-return $r_t = 100\log(P_t/P_{t-1})$, we compute VaR for the filtered

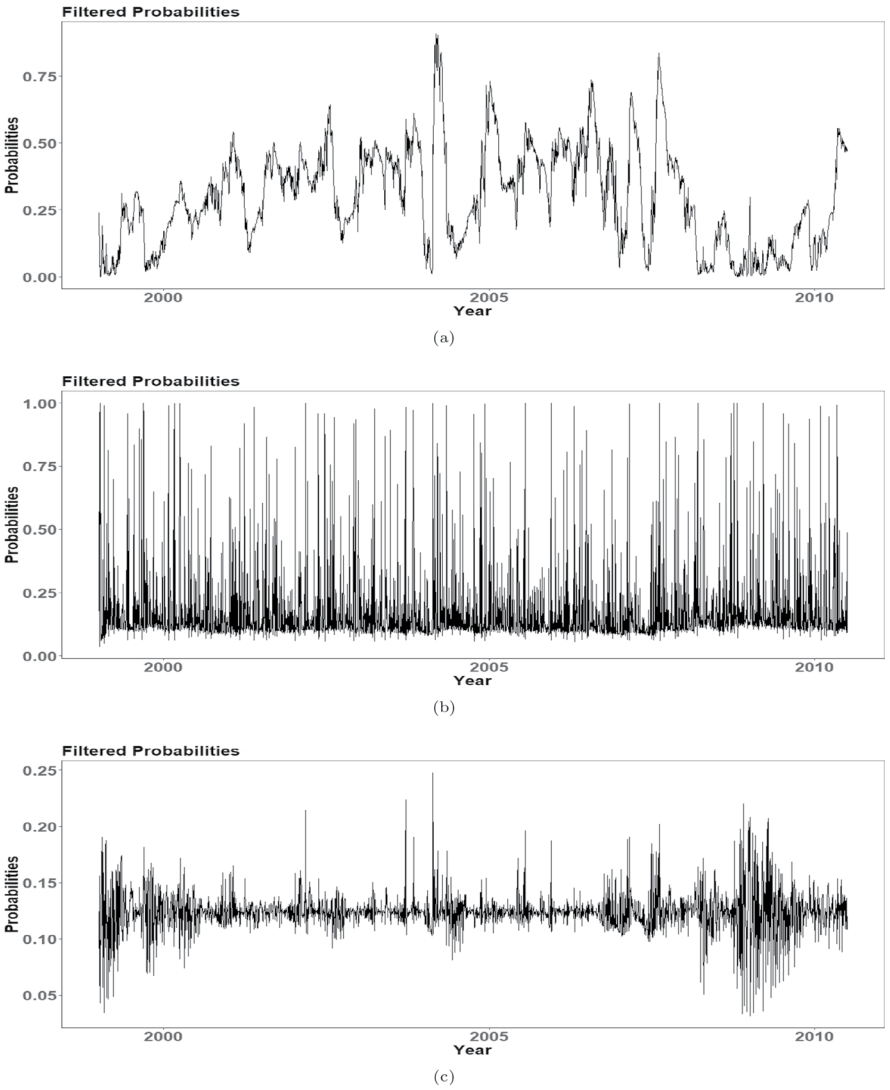


Fig. 6 Estimated probabilities of being in the high volatility regime by the three estimators for the yen, using the first estimation window. The estimators are: **a** robust with $\delta = 0.820$, **b** QML-n, **c** QML-t

series $r_t^* = r_t - (\phi r_{t-1})$, where ϕ is the autoregressive parameter estimated from the model. We estimate ϕ using the 3,000 observations in each window. The next observation used for comparison with the estimated VaR, r_{t+1}, is also filtered using the corresponding estimated value of ϕ.

Table 2 presents the results of the VaR backtesting p-values for $\alpha = 1\%$ and 5% for the indices of the four currencies, according to five estimators, including the robust estimator with three different values of δ. Figure 7 shows the observed returns and VaR estimates from the three methods, with the robust estimator shown for $\delta = 0.820$.

Performance is assessed using four tests based on the sequence and proportion of hits, i.e., cases where losses exceed the VaR. The tests are: (i) the unconditional coverage (UC) test of Kupiec (1995; ii) the independence test of Christoffersen (1998; iii) the conditional coverage (CC) test of Christoffersen (1998); and (iv) the dynamic quantile (DQ) test of Engle and Manganelli (2004), based on a linear regression. For all tests, the null hypothesis is that the model is correctly specified; thus, higher p-values (closer to 1) indicate better performance.

In the analysis of the proportion of hits for the 5% VaR, the observed hit rates for the QML-n and QML-t estimators are close to the nominal level. However, they are consistently above the nominal level for the QML-n estimator and below it for the QML-t estimator. As expected, the QML-n estimator yields a higher number of hits. For the 1% VaR, the proportion of hits for both estimators is below the nominal level for all series, with larger deviations for the pound and yen series.

For the robust method, we expect fewer hits as δ increases, which is indeed observed. An analysis of the observed hit rates and the test results can therefore help guide the choice of the parameter δ.

For the three values considered, $\delta = 0.975$ appears to be the most appropriate, as the observed proportion of hits is closest to the nominal level. However, for the 1% VaR, except for the Swiss franc series, the proportion of hits is higher than the nominal level for $\delta = 0.900$ and lower than the nominal level for $\delta = 0.975$. This may indicate that, for these series, a value of δ between 0.900 and 0.975 could be more appropriate.

Since four statistical tests are applied to four series, there are a total of 16 tests for each VaR level. For both VaR levels, the robust estimator with $\delta = 0.975$ and the QML-t estimator pass the largest number of tests (i.e., yield p-values greater than 0.05): 14 for the 5% level and 13 for the 1% level.

In summary, considering both the proportion of hits and the four statistical tests, the robust estimator with $\delta = 0.975$ and the QML-t estimator exhibit the best performance. Additionally, this exercise illustrates how the value of δ can be calibrated in an applied setting.

6 Concluding remarks

The objective of the paper is to evaluate the performance of several MS-GARCH model estimators in the presence of additive outliers. Since there are no studies on robust estimation techniques for these models, this work contributes a simulation-based performance analysis under different contamination scenarios. The proposed method adapts the robust estimation approach of Boudt et al. (2013) to the Markov regime-switching framework.

Our results can be divided into two parts: the Monte Carlo simulation and the empirical application. The main conclusions from the simulation results are: (i) the QML estimator, assuming a normal conditional density for each regime, performs poorly as the proportion and magnitude of outliers increase, both in terms of filtering accuracy and parameter estimation; (ii) the QML estimator with a Student-*t* distribution with four degrees of freedom as the conditional density for each regime shows

Table 2 Back-testing of the 5% and 1% VaR for the four currencies using the MS(2)-GARCH(1,1) model under different estimation methods

Var	Estimator	δ	%Hit	UC	IND	CC	DQ
Euro							
5%	Rob	0.820	7.042	0.000	**0.246**	0.000	0.000
	Rob	0.900	7.003	0.000	**0.139**	0.000	0.000
	Rob	0.975	5.052	**0.904**	**0.203**	**0.441**	**0.207**
	QML-n		6.238	0.005	**0.069**	0.004	0.008
	QML-t		4.975	**0.953**	**0.173**	**0.394**	**0.785**
1%	Rob	0.820	1.301	**0.139**	**0.344**	**0.214**	**0.497**
	Rob	0.900	1.148	**0.457**	**0.404**	**0.535**	**0.769**
	Rob	0.975	0.498	0.004	**0.718**	0.016	**0.204**
	QML-n		0.880	**0.530**	**0.523**	**0.669**	**0.949**
	QML-t		0.612	0.032	**0.657**	**0.091**	**0.444**
Pound							
5%	Rob	0.820	7.271	0.000	**0.371**	0.000	0.000
	Rob	0.900	7.348	0.000	**0.801**	0.000	0.000
	Rob	0.975	5.396	**0.359**	**0.289**	**0.374**	0.002
	QML-n		6.659	0.000	**0.398**	0.001	0.000
	QML-t		4.746	**0.547**	**0.694**	**0.772**	0.000
1%	Rob	0.820	1.072	**0.716**	0.036	**0.104**	0.000
	Rob	0.900	0.957	**0.823**	0.022	**0.071**	0.000
	Rob	0.975	0.421	0.001	**0.760**	0.003	**0.108**
	QML-n		0.574	0.017	**0.677**	**0.054**	**0.360**
	QML-t		0.536	0.009	**0.698**	0.030	**0.315**
Yen							
5%	Rob	0.820	7.310	0.000	**0.395**	0.000	0.000
	Rob	0.900	7.501	0.000	**0.154**	0.000	0.000
	Rob	0.975	5.243	**0.572**	**0.365**	**0.566**	**0.685**
	QML-n		5.779	**0.074**	**0.923**	**0.203**	**0.287**
	QML-t		4.592	**0.333**	**0.523**	**0.510**	**0.767**
1%	Rob	0.820	1.607	0.004	**0.182**	0.007	0.004
	Rob	0.900	1.684	0.001	0.043	0.001	0.000
	Rob	0.975	0.689	**0.090**	**0.617**	**0.210**	**0.625**
	QML-n		0.536	0.009	**0.698**	0.030	**0.143**
	QML-t		0.651	**0.055**	**0.637**	**0.142**	**0.569**
Swiss Franc							
5%	Rob	0.820	7.348	0.000	**0.179**	0.000	0.000
	Rob	0.900	5.281	**0.513**	**0.174**	**0.320**	0.000
	Rob	0.975	5.166	**0.698**	**0.254**	**0.484**	0.000
	QML-n		5.894	0.041	**0.749**	**0.118**	**0.148**
	QML-t		4.554	**0.289**	**0.062**	**0.100**	0.001

Table 2 (continued)

Var	Estimator	δ	%Hit	UC	IND	CC	DQ
1%	Rob	0.820	1.339	**0.097**	**0.090**	**0.060**	0.000
	Rob	0.900	0.765	**0.209**	**0.144**	**0.156**	0.000
	Rob	0.975	0.765	**0.209**	**0.144**	**0.156**	0.000
	QML-n		0.918	**0.671**	**0.505**	**0.731**	**0.971**
	QML-t		0.689	**0.090**	**0.617**	**0.210**	**0.723**

Moving window of size 3000 starting on 2010-07-02

UC denotes the unconditional coverage test of Kupiec (1995). CC and IND denote, respectively, the conditional coverage and independence tests of Christoffersen (1998). DQ denotes the test of Engle and Manganelli (2004)

P-values greater than 0.05 are shown in bold

reasonable performance in terms of filtering accuracy and in the estimation of several volatility parameters; and iii) the robust estimator also exhibits reasonable performance in both filtering accuracy and parameter estimation.

To evaluate the predictive performance of the estimators, we assess the performance of one-step-ahead VaR forecasts for the exchange rates of four currencies against the US dollar. The main conclusions are as follows: (i) the calibration of δ, which is equivalent to the calibration of $r_{k_{\delta,1}}(u)$, plays an important role; and (ii) the QML estimator with a Student-*t* distribution and the robust estimator with $\delta = 0.975$ deliver the best performance. However, we acknowledge that the empirical study is limited, as it is based on only four exchange-rate series.

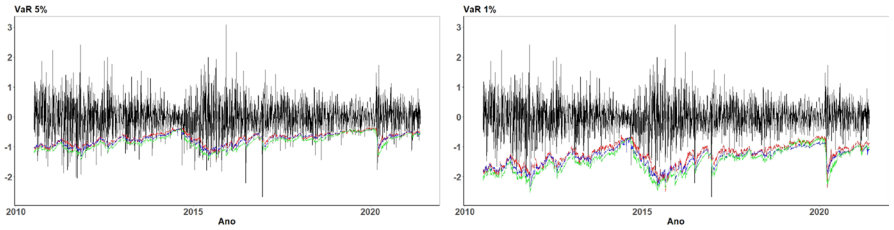
In summary, considering the simulation results and empirical applications, the evidence indicates that the proposed robust estimator is competitive and useful under contamination, but not uniformly superior to QML-t.

We suggest a baseline value given by Eq. (6). As in Muler and Yohai (2008) and Boudt et al. (2013), when introducing a similar procedure, it is common practice to provide a default specification. The final calibration, however, should depend on the practitioner's knowledge of the data under analysis. For instance, if one prefers to overestimate rather than underestimate VaR values, the value of k_δ^* should be increased. Conversely, if it is more important to reduce the influence of possible outliers—even at the risk of falsely identifying them—then smaller values of k_δ^* should be used. Given that the robust estimator is competitive but not uniformly superior to QML-t, we recommend considering both methods in practical applications. The choice between them should also take into account the above considerations regarding the selection of the value of k_δ^*.

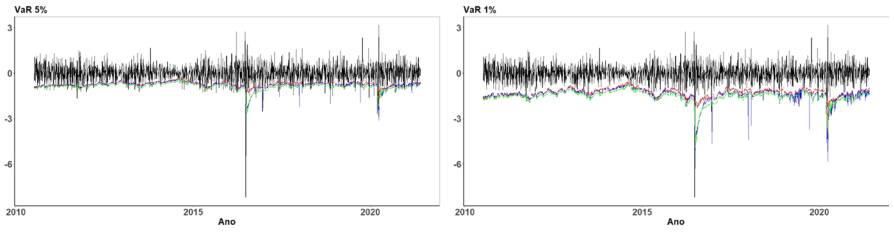
There are several possible extensions of this work, most of which were suggested by the reviewers. For instance, the approach could be extended to other types of outliers and alternative MS-GARCH specifications,while a broader benchmarking study against alternative robust regime-switching methods remains an important direction for future work.

In addition, a more comprehensive empirical analysis across different asset classes should be considered.

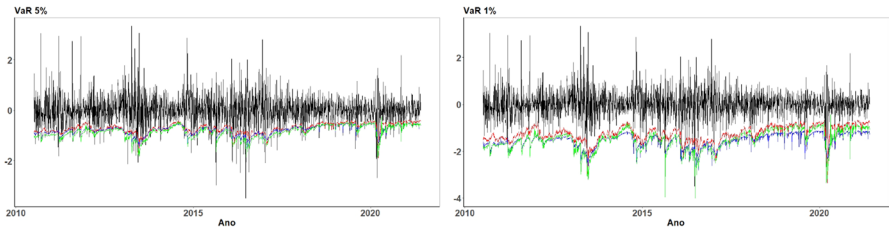
Finally, the derivation of the asymptotic properties of the proposed estimator and a formal analysis of robustness properties, such as breakdown points and influence functions, would be valuable.



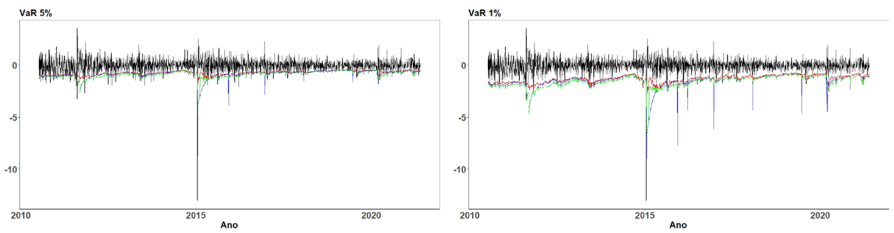
(a) Euro



(b) Libra



(c) Yen



(d) Swiss Franc

Fig. 7 VaR estimate for the four currencies. The black line represents the log return filtered by an AR(1), the line represents the VaR estimated by the robust estimator with $\delta = 0.820$, the blue line by QML-n and the green line by QML-t. VaR 5% is on the left panel and VaR 1% is on the right panel

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Data availability The data are publicly available at https://www.federalreserve.gov/releases/H10/hist/.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval The authors declare that there are no ethical issues.

Consent for publication Not applicable, as the study does not involve human participants and relies solely on financial data.

Consent to participate Not applicable, as the study does not involve human participants and relies solely on financial data.

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